Thesis Progress Report

Student: Muhammad Abdul Ahad

Thesis Title: Hardware Design and Verification of Dilithium – Post-Quantum DSA

Reporting Period: ____________________

Objective

The objective of this thesis is to design a hardware implementation of the Dilithium post-quantum digital signature algorithm and perform its functional verification using a UVM-based verification environment.

Work Completed

• Conducted initial literature review and identified relevant reference implementations.

• Developed a basic understanding of the Dilithium algorithm and its stages.

• Tested the reference software model for use as a golden model.

• Set up GitHub repository and project structure.

• Modified software model to allow custom seed input for controlled testing.

• Uploaded modified software model to repository.

• Configured simulation environment (QuestaSim) and UVM verification framework.

Current Work

• Studying UVM testbench architecture and workflow.

• Compiling and analyzing a reference implementation for hardware design guidance.

Challenges

• Difficulty in fully understanding the Dilithium algorithm.

• Compiler errors while compiling reference implementation.

Planned Work (Next Phase)

• Strengthen understanding of UVM using examples and reference implementation.

• Resolve compilation issues and simulate reference implementation.

• Begin hardware development starting with hashing module.

- Further study hashing and overall algorithm.

Remarks

The project is currently in the foundation and setup phase. Resolving current challenges will enable transition into the hardware design and implementation phase.